

My First Micro336 Assignment

XCM

7/6/2017

This is an extremely simple R Markdown example. You can use the white space to write comments and your answers to any questions that are in the lab manual.

R will execute commands that are between three backticks followed by {r}, like the code block underneath this sentence.

```
#R will not execute this comment because it starts with #, but it will execute the code underneath  
print("Welcome to Micro336!")
```

```
## [1] "Welcome to Micro336!"
```

The preceding three backticks indicate the end of the block of code, so I can put more comments here.

```
# This is my second chunk of R code  
print("I can't wait to learn to use R!")
```

```
## [1] "I can't wait to learn to use R!"
```

```
a<-1  
b<-2  
print(a + b)
```

```
## [1] 3
```

Questions for Assessment:

Answer the following questions for this week's lab homework. Please answer Question #1 prior to Wednesday lab.

- 1) Do you think that water, human, or soil samples will be the most difficult to extract significant quantities of DNA from? Why?
- 2) Fill out the MIMARKS template for the samples you've collected for this week's lab (Lab1-Mimarks.csv):
- 3)
 - a) Did you see bands on your gel for your positive control?
 - b) Did you see a band on your gel for your negative control?
 - c) Did any samples fail to amplify?
 - d) Did PCR band intensity correspond to DNA quantity?
- 4) Based on your own results, do you think a Qubit or a nanodrop is more accurate for quantifying DNA?
- 5)
 - a) Based on entire class Google Doc results, do you think a Qubit or a nanodrop is more accurate for quantifying DNA?
 - b) Based on entire class Google Doc results, are some sample types particularly problematic? Why might this be?

<i>MIMARKS: item</i>	<i>Explanation</i>	<i>Example1</i>	<i>Example2</i>	<i>Sample1</i>	<i>Sample2</i>	<i>Sample3</i>	<i>Sample4</i>
project name	Name of project - you decide	MyMicro336	MyMicro336	NA	NA	NA	NA
investigation type	investigation type	mimarks-specimen	mimarks-specimen	NA	NA	NA	NA
geog. location name	geographic location (country and/or sea,region)	New Zealand	New Zealand	NA	NA	NA	NA
lat lon	geographic location (latitude and longitude)	45.87 S, 170.50 E	45.87 S, 170.50 E	NA	NA	NA	NA
collection date	collection date	1-Jan-18	1-Jan-18	NA	NA	NA	NA
env biome	environment (biome)	grassland	human-associated	NA	NA	NA	NA
env feature	environment (feature)	grassland	human-oral	NA	NA	NA	NA
env material	environment (material)	soil	saliva	NA	NA	NA	NA
env package	environmental package (can be human-associated, soil, or water)	soil	human-associated	NA	NA	NA	NA
target gene	target gene	16S rRNA	16S rRNA	NA	NA	NA	NA
optional metadata 1	Use this column to record other important metadata	soil type = clay		NA	NA	NA	NA
optional metadata 2	Use this column to record other important metadata		time since last tooth-brushing = 6 hrs	NA	NA	NA	NA

<i>source</i>	<i>dilution</i>	<i>nanodrop-ug.ul</i>	<i>nanodrop:260:280</i>	<i>nanodrop:260:230</i>	<i>qubit-ng.ml</i>	<i>qubit-ng.ul</i>
Arm	0	20	100	144	NA	NA
Nose	0	NA	NA	NA	NA	NA
Soil	NA	NA	NA	NA	NA	NA
Clay	0	NA	NA	NA	NA	NA
Topsoil	0	NA	NA	NA	NA	NA
Water	NA	NA	NA	NA	NA	NA
50 ml	0	NA	NA	NA	NA	NA
1000 ml	0	NA	NA	NA	NA	NA
Serial dil 1:	10	NA	NA	NA	NA	NA
Arm / nose	100	NA	NA	NA	NA	NA
Water 50 / 1000	1000	NA	NA	NA	NA	NA
Soil clay / topsoil	10000	NA	NA	NA	NA	NA
Serial dil 2:	10	NA	NA	NA	NA	NA
Arm / nose	100	NA	NA	NA	NA	NA
Water 50 / 1000	1000	NA	NA	NA	NA	NA
Soil clay / topsoil	10000	NA	NA	NA	NA	72